Multivariable Calculus Final Project

Parrie, Keshav, Diego, and Michael

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1 Construction

1.1 Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or $2.5'$ and $3'$, respectively). The equation that we used was

$$\frac{x^2}{225} + \frac{y^2}{225} = \frac{z^2}{900},$$

where $x \in [-30, 30]$, $y \in [-30, 30]$, and $z \in [0, 36]$. Our cone was domain-restricted to only positive z values, as it would be physically very difficult to construct a cone that extended into the ground.

1.2 Construction Process

Our construction got off to a rough start, with our group not acquiring the proper materials until . . . .

2 Calculations

2.1 Theoretical Precise Value

To calculate the theoretical precise value, we will use the triple integral

$$\iiint\limits_Q f(x, y, z)\, dV.$$

To find the bounds, we will use cylindrical coordinates, as Cartesian coordinates tend to end up messy in such calculations. To find the bounds in cylindrical coordinates, we only need to solve for r, as $\theta \in [0, 2\pi]$ and $z \in [0, 36]$, as given by our domain restrictions above.

To find the bounds for r, we first convert the rectangular x and y to cylindrical coordiantes. Following the conversion, our cone can be represented by the equation $\frac{r^2}{900} = \frac{z^2}{1296}$, which we can solve to find the upper bound for r:

$$\frac{r^2}{225} = \frac{z^2}{900} \implies r^2 = \frac{225z^2}{900} \implies r = \sqrt{\frac{225z^2}{900}} = \sqrt{\frac{z^2}{4}} = \frac{z}{2}.$$

Thus, the upper bound for r is $\frac{z}{4}$, and because the lower bound for r is 0, we know that the complete bounds for r are $[0, \frac{z}{4}]$. Thus, the complete bounds, for all variables, are:

$$\theta \in [0, 2\pi]\,,\ z \in [0, 36]\,,\ \text{and } r \in \left[0, \frac{z}{2}\right].$$

Therefore, we can plug in the bounds and integrand for the triple integral and evaluate it:

$$V = \int_0^{2\pi} \int_0^{36} \int_0^{\frac{z}{2}} r\, dr\, dz\, d\theta = \int_0^{2\pi} \int_0^{36} \left[\frac{r^2}{2}\right]_0^{\frac{z}{2}} dz\, d\theta = \int_0^{2\pi} \int_0^{36} \frac{z^2}{8}\, dz\, d\theta$$
$$= \int_0^{2\pi} \left[\frac{z^3}{24}\right]_0^{36} d\theta = 1944 \int_0^{2\pi} d\theta = 1944\,(2\pi) = 3888\pi \approx 12214.5122 \text{ inch}^3.$$

Thus, the theoretical precise volume of our cone is approximately $\boxed{\text{12214.5122 cubic inches}}$ (or approxinately 7.0686 cubic feet).

2.2 Theoretical Approximate Value

To calculate the theoretical approximate value, we will use the Riemann sum

3 Discussion

3.1 Theoretical Precise Value

Our theoretical precise value is logical.

3.2 Theoretical Approximate Value

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